

Data Model Documentation

Overview

The application manages the following core domains:

- **Employee:** Manages employee details and their submitted expenses and travel requests.
- **Expense:** Tracks submitted expenses, their approval status, and reimbursement details.
- **Approval:** Manages the approval process for expenses and travel requests by managers.
- **Manager:** Manages manager details and their approval actions.
- **Finance:** Manages finance details and policy compliance checks.
- **Reimbursement:** Tracks reimbursement details for approved expenses.
- **Policy:** Defines corporate policies and their thresholds.
- **Audit:** Maintains audit history for all actions on expenses and travel requests.
- **Notification:** Manages notifications sent to employees regarding their expenses and travel requests.
- **Role:** Manages roles and their associated permissions.
- **UserRole:** Associates users with their roles.
- **ApprovalThreshold:** Defines approval thresholds for different roles.

Entity Relationship Diagram

```
[Employee] ---- 1:N ---- [Expense]
[Employee] ---- 1:N ---- [Approval]
[Employee] ---- 1:N ---- [Notification]
[Employee] ---- 1:N ---- [UserRole]
[Expense] ---- 1:1 ---- [Approval]
[Expense] ---- 1:1 ---- [Reimbursement]
[Approval] ---- 1:1 ---- [Manager]
[Manager] ---- N:1 ---- [Approval]
[Policy] ---- 1:1 ---- [ApprovalThreshold]
[Role] ---- 1:N ---- [UserRole]
[Role] ---- 1:1 ---- [ApprovalThreshold]
```

Entities

Employee

Purpose: Represents an employee who submits expenses and travel requests.

Table/Collection Name: Employee

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	VARCHAR	Yes	Max 255 chars	Employee name
email	VARCHAR	Yes	Max 255 chars, unique	Employee email
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many Expense (1:N): Inverse via Expense.employee_id
- Has many Approval (1:N): Inverse via Approval.employee_id
- Has many Notification (1:N): Inverse via Notification.employee_id
- Has many UserRole (1:N): Inverse via UserRole.user_id

Validation Rules:

- email must be non-empty and unique across all records

Expense

Purpose: Represents an expense submitted by an employee.

Table/Collection Name: Expense

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to Employee.id	Employee who submitted the expense
amount	DECIMAL	Yes	> 0	Expense amount
description	VARCHAR	No	Max 255 chars	Expense description
receipt_url	VARCHAR	No	Max 500 chars	URL to uploaded receipt

Field Name	Type	Required	Constraints	Description
status	VARCHAR	Yes	'PENDING', 'APPROVED', 'REJECTED'	Expense status
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (1:N): Foreign key employee_id references Employee . id
- Has one Approval (1:1): Inverse via Approval . expense_id
- Has one Reimbursement (1:1): Inverse via Reimbursement . expense_id

Validation Rules:

- amount must be greater than 0
- receipt_url must be a valid URL if provided

Approval

Purpose: Represents the approval process for an expense or travel request.

Table/Collection Name: Approval

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
expense_id	UUID	No	Foreign key to Expense . id	Expense being approved
manager_id	UUID	Yes	Foreign key to Manager . id	Manager approving the expense
status	VARCHAR	Yes	'APPROVED', 'REJECTED'	Approval status
comment	VARCHAR	No	Max 500 chars	Approval comment
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Relationships:

- Belongs to Expense (1:1): Foreign key expense_id references Expense . id
- Belongs to Manager (1:1): Foreign key manager_id references Manager . id

Validation Rules:

- comment must be non-empty if provided

Manager

Purpose: Represents a manager who approves expenses and travel requests.

Table/Collection Name: Manager

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	VARCHAR	Yes	Max 255 chars	Manager name
email	VARCHAR	Yes	Max 255 chars, unique	Manager email
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many Approval (1:N): Inverse via Approval . manager_id

Validation Rules:

- email must be non-empty and unique across all records

Finance

Purpose: Represents a finance team member who validates policy compliance.

Table/Collection Name: Finance

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier

Field Name	Type	Required	Constraints	Description
name	VARCHAR	Yes	Max 255 chars	Finance name
email	VARCHAR	Yes	Max 255 chars, unique	Finance email
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Validation Rules:

- email must be non-empty and unique across all records

Reimbursement

Purpose: Represents the reimbursement details for an approved expense.

Table/Collection Name: Reimbursement

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
expense_id	UUID	Yes	Foreign key to Expense . id	Expense being reimbursed
amount	DECIMAL	Yes	> 0	Reimbursement amount
status	VARCHAR	Yes	'PENDING', 'COMPLETED'	Reimbursement status
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Expense (1:1): Foreign key expense_id references Expense . id

Validation Rules:

- amount must be greater than 0

Policy

Purpose: Defines corporate policies and their thresholds.

Table/Collection Name: Policy

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	VARCHAR	Yes	Max 255 chars	Policy name
description	VARCHAR	No	Max 500 chars	Policy description
threshold	DECIMAL	Yes	> 0	Policy threshold
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Validation Rules:

- threshold must be greater than 0

Audit

Purpose: Maintains audit history for all actions on expenses and travel requests.

Table/Collection Name: Audit

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
entity_id	UUID	Yes	Foreign key to Expense . id or TravelRequest . id	Entity being audited
entity_type	VARCHAR	Yes	'EXPENSE' or 'TRAVEL_REQUEST'	Type of entity
action	VARCHAR	Yes	'SUBMITTED', 'APPROVED', 'REJECTED', etc.	Action performed
details	JSONB	No	JSON object	Audit details

Field Name	Type	Required	Constraints	Description
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time

Validation Rules:

- entity_type must be one of 'EXPENSE' or 'TRAVEL_REQUEST'

Notification

Purpose: Manages notifications sent to employees regarding their expenses and travel requests.

Table/Collection Name: Notification

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to Employee.id	Employee receiving notification
message	VARCHAR	Yes	Max 500 chars	Notification message
status	VARCHAR	Yes	'PENDING', 'SENT'	Notification status
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (1:N): Foreign key employee_id references Employee.id

Validation Rules:

- message must be non-empty

Role

Purpose: Manages roles and their associated permissions.

Table/Collection Name: Role

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	VARCHAR	Yes	Max 255 chars	Role name
description	VARCHAR	No	Max 500 chars	Role description
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Validation Rules:

- name must be non-empty

UserRole

Purpose: Associates users with their roles.

Table/Collection Name: UserRole

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
user_id	UUID	Yes	Foreign key to Employee.id	User associated with the role
role_id	UUID	Yes	Foreign key to Role.id	Role associated with the user
created_at	TIMESTAMP	Yes	Auto-set on create	Record creation time
updated_at	TIMESTAMP	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (1:N): Foreign key user_id references Employee.id
- Belongs to Role (1:N): Foreign key role_id references Role.id

Validation Rules:

- Both user_id and role_id must be non-null and valid

ApprovalThreshold

Purpose: Defines approval thresholds for different roles.

Table/Collection Name: ApprovalThreshold

Fields:

Field Name	Type	Required	Constraints	Description
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| id | UUID | Yes | Primary key, auto-generated | Unique identifier